

Personal data monetisation model in India: Reimagining the contours of the Digital Personal Data Protection Act, 2023 through a market-oriented approach analysis

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Abstract In the digital age, ensuring data privacy has become increasingly difficult as businesses leverage personal data for profit, often at the expense of individual autonomy. This challenge is exacerbated in India by the Digital Personal Data Protection (DPDP) Act, 2023, which adopts a market-oriented approach, prioritising data processing and economic interests over robust privacy safeguards. The Act deepens existing asymmetries in bargaining power, allowing businesses to exploit the commercial value of personal data while individuals receive no direct benefit. In a society where privacy awareness is limited and financial vulnerability is widespread, this imbalance further exposes individuals to the risks of exploitation and surveillance. To address these concerns, this paper proposes a personal data monetisation model designed to strike a balance between commercial interests and consumer rights. By enabling individuals to license their personal data to businesses in exchange for monetary compensation, the model seeks to restore autonomy and ensure that the economic value of data accrues to its rightful owners. The operationalisation of this model, however, raises critical questions, including how to determine data value, protect privacy in a surveillance-driven economy and address the risks of centralising consent management. This paper critically examines these challenges and explores pathways to implement a framework that aligns economic empowerment with the preservation of privacy as a fundamental right, fostering a more equitable digital ecosystem.

KEYWORDS: personal data, Digital Personal Data Protection Act, 2023, monetisation, India

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INTRODUCTION: THE TRICHOTOMY OF CONSUMER BEHAVIOUR, DATA PROCESSING PRACTICES AND DATA PROTECTION REGULATIONS

It is argued that safeguarding an individual's right to data privacy has become increasingly challenging in the digital age. This claim is predicated on three perspectives: (a) consumer; (b) commercial; and (c) legislative. From the consumer perspective, it has been observed that the delay in introducing comprehensive data protection legislations from the inception of widespread data collection practices by early entrants in the digital market created conditions where entities could collect consumers' data without being subject to stringent regulatory oversight.¹ Consequently, the absence of stringent oversight in terms of clear standards or rules created circumstances for entities to draft privacy policies riddled with lengthy technical legal jargon that was difficult for the average consumer to fully comprehend, resulting in excessive and intrusive data collection practices.² These policies were based on three assumptions: (a) individuals have the capacity to make the correct, appropriate and suitable choice;³ (b) individuals possess the required information to make that choice;⁴ and (c) individuals have adequate time to make that choice.⁵

However, research suggests that consumers predominantly exhibited three behavioural patterns when consenting to data processing.⁶ These patterns are: (a) *bounded rationality*, where the decision-making process or rationality of individuals is constrained by time, resources and cognitive constraints; (b) *heuristic familiarity*, where individuals tend to opt for familiar responses due to positive responses previously received; and (c) *default inertia*, which identifies the reluctance of

individuals to change the default settings they are presented with, whereby these default settings have the possibility of harming consumers by invading their data privacy through collecting excessive amounts of personal data (also known as dark patterns).⁷ This results in consumers consenting to data collection practices without completely understanding all the ramifications. Consequently, these uninformed disclosure practices, in turn, perpetuate information and power asymmetries, resulting in individuals becoming increasingly 'transparent' or 'legible' to those processing personal data, while the processing practices become increasingly opaque.

From a commercial perspective, while newer data protection laws have laid down stricter standards for data collection, early entrants into digital markets have already amassed extensive data on individuals.⁸ This was highlighted in controversies such as the 2016 Google, Amazon, Facebook, and Apple (GAFA) case, where the US bipartisan committee highlighted how these digital giants were able to leverage their early entry into the digital market to collect vast amounts of consumer data and establish dominant positions to significantly influence the digital market landscape.⁹ Their early access enabled them to amass detailed user profiles and develop comprehensive market intelligence, resulting in anti-competitive conduct and reduced consumer privacy options.¹⁰ For instance, Facebook was able to create robust user networks through the data it amassed and used this intelligence to carry out 'killer acquisitions' of upcoming competitors.¹¹ As a result, consumers had fewer viable alternatives and had to consent to data-sharing terms and conditions that prioritised the companies' commercial

interests over the consumers' privacy interests.¹² Therefore, the early entry of the GAFA giants, coupled with their intrusive data aggregation strategies, resulted in weaker privacy standards, compelling consumers to comply with expansive data collection practices to access and use essential digital services.

Other controversies include the 2018 policy of the Ministry of Road Transport & Highways (MoRTH) in India.¹³ This policy granted entities access to its Sarathi and m-Vahan databases, which contained individuals' vehicular data.¹⁴ Although this policy acknowledged the possibility of triangulation (linking of datasets), it did not implement tangible measures to prevent it.¹⁵ Eventually, this policy was scrapped in 2020; however, its scrapping did not necessitate the erasure of the datasets these entities procured and possessed.

From a legislative perspective, there are two considerations to be noted. Firstly, there is currently no requirement for entities under existing data protection legislations to erase previously collected consumer data.¹⁶ In other words, data protection laws do not apply retrospectively and entities are not obligated to obtain fresh consent for processing the data that is already in their possession, nor are they required to delete it.¹⁷ Secondly, current legislation does not necessitate that consumers should be granted access to essential digital services even if consent to data processing is denied by the consumer.¹⁸ In other words, access continues to remain contingent upon the consumer providing their consent to data processing. This means that opting out of data processing is generally not an option if one wishes to access digital content, articles, resources, etc. As a result, consent may not be entirely voluntary, as consumers may agree to the terms and conditions to use these services. Thus, these three perspectives highlight the obscure nature of privacy in the digital age and the challenges associated with ensuring the same, despite subsequent regulatory developments.

THE DIGITAL PERSONAL DATA PROTECTION (DPDP) ACT 2023: UNCOVERING ITS TILT TOWARDS MARKET-ORIENTED APPROACH FOR PERSONAL DATA MONETISATION

In the realm of data regulation, there has been considerable discourse surrounding the concept of duality of data. On the one hand, there exists a rights-based approach that is anchored in the fundamental right to data privacy and protection.¹⁹ This approach interprets personal data in terms of individual rights, emphasising the need to safeguard and control the movement of personal data.²⁰ An example of a legal framework based on such a rights-based approach is the European Union General Data Protection Regulation (EU GDPR).²¹ Such legal frameworks are generally characterised by stringent regulations and standards to protect the data privacy of individuals.²²

On the other hand, the market-oriented approach views personal data as an economic commodity, asset or input for goods and services.²³ This approach is evident in legal frameworks such as the 2013 Organization for Economic Co-operation and Development (OECD) Privacy Guidelines²⁴ and Asia-Pacific Economic Co-operation (APEC) Privacy Framework,²⁵ which exemplify the market-oriented approach by focusing on personal data as a valuable resource in the global market.²⁶ For instance, the OECD Privacy Guidelines emphasise the importance of promoting an environment conducive to cross-border personal data transfer and harmonisation of data protection regulations.²⁷ This indicates that the Guidelines attempt to focus on enabling the seamless movement of personal data for business and economic purposes and minimising barriers to personal data transfer to facilitate smoother market operations for businesses operating across different jurisdictions.²⁸ The above analysis emphasises the significance of global personal data exchange in the context of international trade and commerce, whereby personal

data becomes an economic commodity for businesses.

Multiple states, however, have introduced legislation harmonising market needs with the increasing demand to protect the right to privacy (rights-based approach). This means that the status quo is to strike a balance between the market-oriented approach and the rights-based approach, with an evident trend towards prioritising the protection of citizens' privacy.²⁹ By emphasising the rights-based approach, the data protection regulations aim to ensure that personal data usage by businesses does not violate individuals' data privacy rights,³⁰ where this is likely to mitigate the risks of exploitation or misuse of personal data. For instance, the EU GDPR, which stands as one of the most stringent data protection regulations globally, is heavily tilted towards a rights-based approach while still retaining the elements of a market-oriented approach. The GDPR places an individual's right to data privacy at the forefront by implementing data protection through design and default,³¹ meaning that it integrates data protection and privacy measures into the design of systems, products and services from the outset of their creation or manufacture.³²

Further, in the event that the provisions of the GDPR are violated by businesses, strict penalties are imposed, ranging from €10m to imprisonment, to ensure compliance.³³ At the same time, the GDPR has elements of a market-oriented approach because: (a) it allows cross-border personal data transfer,³⁴ empowering businesses to transfer personal data for their internal business purposes; (b) it gives data portability rights,³⁵ empowering consumers to transfer their personal data between service providers and enhancing business competition; (c) it reduces the compliance burdens for personal data transfer,³⁶ incentivising businesses to facilitate greater personal data transfer; and (d) it imposes fines proportional to the company's revenue,³⁷ ensuring that large businesses

adhere stringently to the provisions of the GDPR.

Considering the foregoing observations, it is argued that the challenge of upholding consumers' right to data privacy in the digital age is likely to be exacerbated by the enactment of the DPDP Act. This largely stems from the premise that the DPDP Act is tilted more towards a market-oriented framework rather than a rights-based approach, which is supported by various aspects of the DPDP Act as follows.

First, this shift becomes more evident through the exclusion of the term 'privacy', a foundational concept of all data protection regulations.³⁸ Although the DPDP Act attempts to safeguard the data privacy rights of individuals by giving them various rights — such as the right to give express consent and withdraw it,³⁹ right to correct and erase personal data,⁴⁰ right of grievance redressal⁴¹ and right to nominate⁴² — it could be argued that the Act departs from a rights-based framework, evidenced by the omission of the term 'privacy'. The exclusion of privacy, a concept that is intrinsically linked to data protection, as noted above, raises issues pertaining to the Act's real objective. Unlike the DPDP Act's draft form, ie the Personal Data Protection Bill, 2019 (PDP Bill), which categorically mentions privacy as the guiding principle of the legislation, the DPDP Act's silence on this principle may suggest a shift in its focus.⁴³ This omission could suggest that the Act is less concerned about safeguarding individual privacy and more focused on facilitating the creation of a regulatory environment that is conducive to data-driven markets.

Secondly, a comparative analysis with the DPDP Act's draft form, ie the PDP Bill, may also suggest its market-oriented tilt. This suggestion may be supported by two claims:

- *Broader protection to data principals:* The PDP Bill was more comprehensive in nature and offered broader rights and protections to data principals in alignment

with global data protection standards. This was evident in the extensive provisions of the Bill, which aimed to proactively ensure privacy and protection for data principals. One notable example was the privacy by design policy, which the PDP Bill mandated for every data fiduciary.⁴⁴ This policy required fiduciaries to outline technical and organisational measures to anticipate, identify and mitigate harm to data principals.⁴⁵ It included provisions such as the obligations of data fiduciaries to protect privacy throughout the data life cycle, the use of certified technology standards for secure data processing and the enforcement of end-to-end data protection, ensuring that privacy is maintained from the point of data collection to its deletion.⁴⁶

This concept of privacy by design is recognised as a cornerstone of data protection law globally.⁴⁷ For instance, the GDPR enshrines it as a core principle under Article 25.⁴⁸ The GDPR requires organisations to integrate data protection measures into the development and operation of their processing activities from the outset, ensuring privacy is embedded as a default setting rather than treated as an afterthought.⁴⁹ This principle underscores a proactive and preventive approach, reflecting a rights-based framework that prioritises the autonomy and security of individuals.

The DPDP Act omits any explicit reference to a privacy-by-design policy, potentially signalling a significant departure from the rights-based emphasis of the PDP Bill. This omission may indicate a shift in the priority and focus of the legislation. Without mandating the systemic integration of privacy safeguards into data processing systems from the outset, the DPDP Act appears to embrace a reactive and flexible compliance framework. Such a framework can be argued to prioritise adaptability and operational simplicity for businesses,

offering flexibility to comply with data protection obligations as issues arise rather than embedding comprehensive protections by default from the outset.

- *Rights of data principals:* The PDP Bill represented a rights-based approach to data protection, granting individuals comprehensive control over their personal data. Key rights, such as the right to data portability and the right to be forgotten, were included as explicit provisions designed to empower data principals.⁵⁰ These rights are aligned with global data protection standards, emphasising individual autonomy and accountability for data fiduciaries;⁵¹ however, the DPDP Act supplemented by the Draft DPDP Rules reflects a significant departure from this framework. The Draft DPDP Rules operationalise the DPDP Act to safeguard data principals' personal data rights. With provisions for informed consent, data erasure and grievance redressal, the Draft DPDP Rules empower data principals and strike a balance between regulation and ease of doing business. Under the PDP Bill, the right to data portability enabled individuals to transfer their personal data between service providers in a structured, machine-readable format.⁵² This right empowered consumers to move freely between digital platforms, reducing dependency on specific providers while fostering competition and innovation in the digital economy.⁵³ Its absence in the DPDP Act may potentially underscore lesser rights available to data principals, limiting their control over how and where their data is utilised.

The DPDP Act does provide for the erasure of personal data, which superficially resembles the right to be forgotten, yet it lacks explicit recognition and procedural safeguards that define rights under the PDP Bill. The right to be forgotten, as understood in global frameworks such as the GDPR, goes beyond erasure.⁵⁴ It enables individuals to

request the removal of data from public access when it is no longer necessary for processing or when consent is withdrawn.⁵⁵ Importantly, it incorporates mechanisms for balancing individual privacy with public interests, such as freedom of expression.⁵⁶ The DPDP Act's silence on such balancing mechanisms and its omission of adjudicatory provisions for public data suggest a more constrained understanding of privacy protection.

The Draft DPDP Rules further emphasise the procedural aspects of grievance redressal and outline the obligations of data fiduciaries. For example, fiduciaries must provide transparent mechanisms for data principals to exercise their rights, such as withdrawing consent or correcting inaccuracies in their data.⁵⁷ While these provisions offer some avenues for data principal engagement, they lack the depth and breadth of protections envisaged under the PDP Bill. The Draft DPDP Rules do not expand on or reintroduce broader rights such as data portability or a fully articulated right to be forgotten, potentially reinforcing the perception that the DPDP framework prioritises streamlined compliance for businesses over comprehensive individual rights.

The omission of expansive rights and the narrower scope of protections under the DPDP Act reflects a shift toward a market-oriented framework. By reducing the regulatory burden on businesses and simplifying compliance requirements, the Act appears to prioritise economic utility over individual empowerment. The Draft DPDP Rules support this orientation by focusing on procedural efficiency for fiduciaries, such as standardised notice and consent mechanisms, rather than expanding substantive rights for data principals.⁵⁸

This market-oriented approach may facilitate innovation and reduce costs for businesses, but it comes at the expense of

the proactive and rights-based safeguards that were central to the PDP Bill. As a result, the DPDP framework represents a fundamental recalibration of priorities, where data protection is framed less as a tool for individual empowerment and more as an enabler of economic activity within a regulated digital ecosystem.

Thirdly, Section 6(5) of the DPDP Act assigns the consequences of consent withdrawal to the data principal, allowing data fiduciaries to restrict or deny services if consent is withdrawn.⁵⁹ This provision can be interpreted as reflective of a regulatory framework that prioritises business continuity and operational ease, particularly when viewed alongside international and prior domestic data protection frameworks such as the GDPR and the PDP Bill. Under the GDPR, Article 7(3) emphasises that withdrawing consent should be as easy as giving it and should not result in detriment to the data subject.⁶⁰ While the GDPR does not entirely preclude the possibility of service limitations due to consent withdrawal, it encourages data controllers to design systems that are proportionate and do not unfairly penalise individuals for exercising their rights.⁶¹ For instance, the GDPR appears to strike a balance by promoting mechanisms that allow for partial or alternative access to services, particularly when consent withdrawal affects only specific data processing activities.⁶²

Similarly, the PDP Bill, as a precursor to the DPDP Act, contained provisions that underscored the fiduciary's accountability in managing consent withdrawal.⁶³ The Bill included safeguards, such as transparency requirements and data protection impact assessments (DPIA), particularly for significant fiduciaries, which could encourage mitigation of service disruption when consent is withdrawn.⁶⁴ While the PDP Bill did not explicitly mandate partial service access, its emphasis on fiduciary accountability may have created

an expectation that reasonable adjustments should be considered to uphold data principal rights.

Section 6(5) of the DPDP Act, however, does not expressly require fiduciaries to make such adjustments.⁶⁵ The provision appears to allow data fiduciaries the discretion to discontinue or restrict access to their services upon consent withdrawal without an obligation to explore alternatives.⁶⁶ This could suggest an approach that places greater emphasis on ensuring service continuity and reducing operational burdens for fiduciaries. As a result, this design may be interpreted as fostering flexibility for businesses to define their service models while allowing them to maintain streamlined compliance processes. This framework could also be viewed as encouraging a more market-oriented regulatory stance. By not mandating fiduciaries to accommodate partial access or implement adjustments, the provision aims to minimise compliance and operational costs for businesses. This stance, however, might result in limited flexibility for data principals to exercise their right to withdraw consent without potentially losing access to services they consider essential.

Fourthly, Section 3 of the DPDP Act, establishes the scope of the Act, outlining its applicability and exclusions, with Section 3(c)(ii) addressing the treatment of publicly available data.⁶⁷ While the Act's territorial applicability mirrors global frameworks such as the GDPR — extending its reach to personal data processing within India and by entities outside India offering goods or services to Indian data principals — it departs significantly from international best practices regarding the processing of publicly available personal data.⁶⁸ Under the GDPR, while the territorial scope is similarly broad, the framework imposes strict principles such as purpose limitation under Article 5(1)(b) and ensures that all personal data, including publicly available information, is processed lawfully, fairly and transparently.⁶⁹ Importantly, the GDPR mandates that

consent must be explicitly obtained for each specific purpose of processing, even for data disclosed in the public domain.⁷⁰

By contrast, Section 3(c)(ii) of the DPDP Act exempts publicly available data from its protections, allowing such data to be processed without explicit consent or safeguards such as purpose limitation.⁷¹ This exemption implies that data made public — whether voluntarily by the data principal or under a legal obligation — can be freely used by data fiduciaries and other entities. For instance, if an entrepreneur's contact details are published as part of mandatory corporate filings, such data can be processed without the individual's consent, leaving them vulnerable to unsolicited calls, messages and e-mails. Beyond corporate scenarios, this provision also exposes individuals who have disclosed sensitive personal information online, such as their sexuality, race, political beliefs or other identifying factors, to significant risks. Many people share such information on social media platforms like X (formerly Twitter,) Reddit or Facebook for specific purposes, often to build communities, seek support or advocate for causes.

The blanket exemption for publicly available data under the DPDP Act allows third parties to harvest this information and potentially misuse it, such as by targeting individuals with discriminatory profiling, harassment or exploitative practices. This lack of protection contrasts sharply with global privacy norms, where publicly available data is still subject to stringent safeguards.⁷² For example, under the GDPR, even data disclosed publicly cannot be processed unless it aligns with one of the six legal bases outlined in Article 6.⁷³ Furthermore, each processing purpose must be specified, legitimate and accompanied by express consent.⁷⁴ The failure to justify the processing of publicly available data under such standards has led to significant consequences elsewhere.⁷⁵ In Italy, for instance, the inability of large language

models (LLMs) like ChatGPT to justify the processing of publicly available personal data without explicit consent or purpose specificity resulted in their temporary ban.⁷⁶ This highlights the importance of treating publicly available data with care, ensuring that individuals retain control over how their information is used, even when it enters the public domain.

The DPDP Act's approach not only raises concerns about privacy risks but also conflicts with principles of informational autonomy and purpose limitation, which are central to global data protection frameworks and India's own constitutional jurisprudence following the Puttaswamy judgment. Moreover, the Act's lack of retrospective application exacerbates these risks, leaving data disclosed prior to its enactment unprotected and vulnerable to exploitation.

Fifthly, Section 7(a) of the DPDP Act provides that a data fiduciary may process personal data for a specified purpose if the data principal has voluntarily provided such data and has not expressly refused consent for its use.⁷⁷ This provision appears to adopt a less restrictive approach to consent, potentially facilitating the flow of personal data in the market. Unlike data protection regimes such as the GDPR, which generally mandate express, informed and purpose-specific consent, the DPDP Act allows for processing based on implied consent, provided the data principal has not actively objected to the use of their data.⁷⁸ This could simplify compliance for data fiduciaries by removing the need to seek fresh and explicit consent for each instance of data processing, particularly when the initial consent is broad enough to accommodate multiple specified purposes.

This approach, when coupled with the exemption of publicly available personal data under Section 3(c)(ii), may further encourage the reuse of data without additional consent requirements.⁷⁹ Publicly available data, whether shared voluntarily by individuals or disclosed due to legal

obligations, is excluded from the scope of the Act. Consequently, data fiduciaries may not be bound by the usual safeguards, such as purpose limitation or transparency obligations, when processing such data. For instance, individuals who publicly disclose sensitive personal information about their sexuality, race or other identifying factors on platforms like X (formerly Twitter) or Reddit may not have anticipated that their data could be repurposed by third parties, such as for profiling or targeted advertising. Without explicit consent requirements, such processing could raise concerns about the adequacy of protections for informational autonomy, particularly in cases where the data was shared with specific intentions or within specific contexts.

Sixthly, Section 17(3) of the DPDP Act provides the central government with the discretionary power to exempt specific data fiduciaries, including start-ups, from certain obligations under the Act, taking into account the volume and nature of personal data being processed.⁸⁰ The exemptions may extend to requirements such as issuing consent notices, conducting DPIAs, performing periodic audits, appointing data protection officers (DPOs) and providing requested information to data principals.⁸¹ This provision appears to reflect a market-oriented approach aimed at reducing compliance burdens on start-ups and smaller entities, enabling them to focus on innovation and scalability without being weighed down by extensive regulatory requirements.

When contrasted with global frameworks such as the GDPR, the market orientation of the DPDP Act becomes more evident. Under the GDPR, there is no explicit exemption for start-ups or small entities from fundamental obligations such as conducting DPIAs, appointing DPOs or responding to data subject requests. While the GDPR does provide proportionality mechanisms, such as risk-based assessments, it largely maintains uniform obligations to ensure accountability

and protect individual rights.⁸² Similarly, the PDP Bill did not explicitly provide for such exemptions, emphasising a more privacy-centric approach that prioritised individual rights and accountability across all data fiduciaries, irrespective of size or scale.⁸³

The DPDP Act's exemption framework may result in increased flexibility for start-ups, potentially fostering innovation and market growth. It may also lead to uneven levels of data protection, however, as smaller entities processing sensitive personal data could be exempted from critical safeguards such as DPIAs or periodic audits. This raises concerns about whether such exemptions could inadvertently compromise the privacy rights of individuals and reduce accountability in data processing practices. Moreover, exempting entities from obligations to provide information to data principals might limit transparency and restrict individuals' ability to exercise their rights effectively.

While the GDPR and PDP Bill approach data protection from a rights-based framework, prioritising uniform accountability mechanisms to protect individuals' informational autonomy, the DPDP Act appears to lean toward a more market-friendly model. By focusing on reducing regulatory burdens for specific entities, the Act places an emphasis on promoting ease of doing business. This orientation is reflected in its overarching intention to promote data processing and economic utility over robust data privacy safeguards. The absence of explicit recognition of 'privacy' as a foundational principle, as seen in its predecessor legislation, and the prioritisation of simplified compliance mechanisms suggest a shift in focus. Provisions such as Section 3(c)(ii), which exempts publicly available data from protection, and Section 7(a), which allows implied consent for processing, emphasise operational flexibility and facilitate the free flow of data, often at the potential cost of individual informational autonomy.

Additionally, the exemption framework under Section 17(3), which relieves specific entities such as start-ups from critical obligations, further highlights the Act's market-centric emphasis. This emphasis becomes even more apparent upon contrasting it with global data protection frameworks such as the GDPR and even the PDP Bill, which adopt a rights-based approach that prioritises individual autonomy, purpose limitation and robust safeguards for personal data, including mechanisms such as the right to data portability, privacy by design and explicit consent requirements. In the context of India's growing digital economy, the DPDP Act's market-oriented approach could serve as a foundation to enable data monetisation and enhance economic activities.

In light of the aforementioned arguments, it may be reasoned that, in the context of India, the enactment of the DPDP Act — a prospective legislation which may be interpreted as incorporating certain market-oriented elements, as discussed above — might achieve only limited success in addressing data privacy concerns and reducing bargaining power asymmetries. This underscores the potential need to explore frameworks that could enhance individual autonomy and control over personal data while balancing rights-based and market-oriented approaches.

One such possibility could involve the development of a personal data monetisation model tailored to the Indian legal and socio-economic context. This model may provide individuals with the option to commercialise their personal data through structured, transparent mechanisms, leveraging digital infrastructure such as the India Stack in conjunction with the provisions of the DPDP Act, the Draft DPDP Rules, and the India Data Accessibility and Use Policy, 2022. The framework could, therefore, aim to redefine the flow of value within the data economy: instead of businesses deriving unilateral commercial benefits

from consumer data, individuals themselves may potentially receive direct economic compensation for sharing their data.

Unlike existing practices, which primarily enable the flow of non-personal data from government repositories to businesses in exchange for payments to the state, this model could envision a consumer-centric system where businesses directly compensate individuals. Such an approach may offer relatively greater control, autonomy and self-determination to users by empowering them to make informed choices about the use of their personal data. While the model could contribute to mitigating the bargaining power asymmetry between businesses and consumers, it also raises critical concerns about privacy trade-offs, consent vulnerabilities and potential misuse of data in a society where awareness about privacy and data rights remains limited.

Accordingly, the implementation of such a model must carefully address the unique challenges associated with monetising personal data. These include ensuring informed consent, establishing robust safeguards to prevent misuse, addressing power imbalances in pricing negotiations and recognising the fluid value of personal data across contexts and time. Further, it would be essential to consider whether the proposed infrastructure could inadvertently centralise data collection and exacerbate surveillance risks. By addressing these concerns, the model could offer a novel yet cautious approach to integrating personal data monetisation into India's digital governance framework.

ESTABLISHING PROPERTY RIGHTS IN PERSONAL DATA FOR ENABLING MONETISATION OF PERSONAL DATA: A LEGAL IMPERATIVE

Since the 1970s, there has been considerable debate around property rights in data.⁸⁴ This has evolved from possessing property rights in tangible assets (like physical property) to

intangible assets (like ideas and creations) to safeguard the interest of property owners such as consumers and other individuals in the context of intellectual property.⁸⁵ This leads to the necessity to establish property rights in personal data, which mainly germinates from the economic value of personal data to businesses and the potential impact on individuals⁸⁶ in the following manner:

- a. If property rights are not assigned to personal data, it legitimises the actions of businesses to appropriate the personal data and exploit the individuals to whom it belongs, as this personal data significantly assists businesses in developing better products and services by predicting consumer trends, tastes and preferences.⁸⁷
- b. The inability to assign property rights in personal data, in turn, renders individuals to whom it belongs vulnerable to massive corporate power in terms of asymmetrical bargaining power and erodes the individuals' autonomy, control, privacy and right to informational self-determination.⁸⁸

Thus, the debate begins to revolve not around whether property rights in personal data should exist but around who should be the rightful holder of such rights: individuals to whom the personal data belongs or organisations that utilise this data. It is argued that individuals must be granted property rights in their personal data because it empowers them to dictate the extent of data processing and may even enable them to deny access to it.⁸⁹ For instance, in the US, there has been a stronger emphasis on establishing property rights in personal data. One notable example is Andrew Yang's campaign, which advocated for individual ownership of digital personal data.⁹⁰ Additionally, Senator John Kennedy introduced the 'Own Your Own Data' Act in 2019, proposing that each individual possesses exclusive property rights to their digital personal data.⁹¹

When these property rights in personal data are taken a step beyond by contemplating monetisation of personal data, it is evident that there have been similar global efforts in the past. In 2020, California's Governor, Gavin Newsom, explored the concept of a 'data dividend' for California residents, aiming to ensure that individuals receive a portion of the profits generated from their personal data.⁹² Thus, this combination of property rights and monetisation of personal data shows the evolving landscape of data governance by advocating for a hybrid regime (personal data monetisation model) to reconcile the dual nature of personal data.⁹³ In other words, the evolving data protection landscape seeks to protect the fundamental right to data privacy (rights-based approach) and simultaneously allow individuals to benefit from economic opportunities associated with their personal data as an economic commodity or tradeable asset (market-oriented approach). Therefore, the market-oriented approach to personal data is rooted in the concept of property rights in personal data, as it empowers individuals to actively participate in the economic utilisation of their personal data and exercise greater control over their assets.⁹⁴

LEVERAGING INDIAN COPYRIGHT LAW FRAMEWORK FOR PERSONAL DATA MONETISATION

The realm of digital data represents a significant facet of every individual's identity and expression because it allows individuals to express, communicate and transact.⁹⁵ This means that data inherently encapsulates individuals' intrinsic values of speech and expression, ie fundamental rights as given under the Indian Constitution.⁹⁶ Examples of such data include literary work, songs, computer programs and any other work that the individual has created or reproduced by monetising their fundamental right to speech and expression.⁹⁷

In India, such data is protected under the copyright framework, ie the Copyright Act, 1957 (1957 Act).⁹⁸ In the context of data monetisation, the Indian copyright framework serves as an apt example to demonstrate how data may be generally monetised.⁹⁹ Under Chapter VI of the 1957 Act, copyright owners are allowed to grant any 'interest' in their works to other parties by way of licensing.¹⁰⁰ This 'interest' in the copyrighted work may be subject to certain restrictions as may be agreed between the copyright owner (licensor) and the other parties (licensees). The purpose of licensing is to primarily earn money. This may be done in two ways: first, the copyright owner may receive a lump-sum one-time payment, or secondly, the copyright owner may be paid royalties, which may be mutually decided between the copyright owner and the other parties. These royalties may be a fixed percentage on each sale made with respect to the copyrighted work, a fixed percentage on monthly sales, or any other arrangement as may be mutually decided.

Similarly, in the context of the DPDP Act, the Indian copyright framework allows for reimagining the fundamental right to data privacy and protection by showing how a similar monetisation framework may be applied to personal data. Under this system, consumers would be able to monetise their personal data by licensing it to businesses. This would mean granting businesses a specific 'interest' in their personal data under clearly defined terms. Crucially, consumers would retain ownership of their personal data and exercise greater autonomy, control and the right to informational self-determination.

To operationalise this model, the concept of data trust becomes pivotal. A data trust is a legal structure where individuals pool their data under the stewardship of a neutral entity. This trust is tasked with managing access, ensuring compliance with regulatory norms and protecting the rights of data contributors. Data trusts align well with the principles of data justice, emphasising

inclusivity, accountability and public benefit. By acting as intermediaries, data trusts can ensure that the monetisation of personal data does not disproportionately benefit businesses at the expense of individual autonomy. Additionally, they can foster transparency by clearly outlining how data is used, shared and monetised.

It is noteworthy to discuss the types of personal data that will be subject to the personal data monetisation model in India. Section 3 of the DPDP Act discusses the types of personal data that will be subject to the provisions of the Act.¹⁰¹ Accordingly, the personal data monetisation model will be applicable and non-applicable to the following types of personal data:

Applicable to personal data processed in India: The personal data monetisation model will be applicable to the personal data of data principals that are processed in India, provided such personal data is available in digitised format.¹⁰² This means that the model applies to both citizens and non-citizens and residents and non-residents, provided their personal data is processed in India in compliance with the provisions of the DPDP Act.

- *Applicable to personal data processed outside India:* The personal data monetisation model will be applicable to the personal data of data principals that are processed outside India, where such processing is connected with any goods or services offered to the data principal in India.¹⁰³ This means that the model applies to both non-citizens and non-residents, provided their personal data is processed outside India, and such processing is for any goods or services that they have opted for or availed in India.
- *Non-applicable for processing for non-business purposes:* The personal data monetisation model will not be applicable to personal data that is processed by any individual for non-business purposes, such as personal or domestic.¹⁰⁴

- *Non-applicable for publicly available personal data:* The personal data monetisation model will not be applicable to personal data that is publicly available.¹⁰⁵

INCENTIVE ANALYSIS FOR PERSONAL DATA MONETISATION IN INDIA: GOVERNMENT, BUSINESS AND CONSUMER PERSPECTIVES

After having discussed the foundational monetisation framework to monetise personal data in India through the personal data monetisation model, it becomes imperative to discuss the incentives for governments, businesses and consumers to agree to such a unique model. This analysis can be done in a trifold manner as follows:

1. When it comes to the Indian Government's incentives to implement this personal data monetisation model, incentives exist at both the state and central levels in the following manner. First, if the personal data monetisation model is implemented in India, it would allow the Government to levy central-level and state-level taxes on the transactions undertaken between the data principal (consumer) and the data fiduciary (business). This means that when the data principal's personal data is shared and sold to the data fiduciary by the data principal, taxes can be levied on the compensation amount that would be transferred to the data principal by the data fiduciary. Secondly, the Parliament would not be required to amend the entire DPDP Act to better protect the rights of the consumers as, now, the Parliament can directly operationalise the personal data monetisation model under the DPDP Act without amending it. This would save human resources and time invested in the possibility of amending entire legislation and would allow the Parliament to deliberate on other important Bills and issues.

Thirdly, the Government will be able to benefit from a political angle because it would lead to increased buy-in from the voter bank (consumers) as the consumers are likely to be substantially satisfied under this personal data monetisation model due to their increased ability to earn money and profit by selling their personal data.

Fourthly, there is likely to be substantial economic growth. This is because when the data principals (consumers) sell their personal data to the data fiduciaries (businesses), the consumers earn money. This leads to an increase in the disposable income of the consumers, ie the amount and ability of consumers to spend on goods and services. This results in an increase in India's gross domestic product (GDP), because consumers spend more on goods and services, businesses earn more revenue, consumer demand increases and businesses increase production, which leads to a rise in the value of goods and services. This increased GDP results in increased national income, ie the total amount of money earned in India, because businesses earn more revenue, leading, in turn, to higher profits, wages, salaries and investment income. This benefits the Government because: (a) it is able to earn higher tax revenue without increasing the tax rates as more taxes are collected on higher revenues, profits and sales; (b) it reduces social spending, ie the need for the Government to spend on social welfare programmes thanks to increased hiring, wages and salaries; and (c) it creates more fiscal space for the Government due to reduced social spending because, now, the Government can reallocate its resources and alternatively invest in other areas such as green energy, artificial intelligence (AI), nuclear power etc., to stimulate the country's economic growth.

Lastly, personal data monetisation is not expressly prohibited by law in

India. Section 4 of the DPDP Act states that personal data can be processed for a lawful purpose, ie any purpose that is not expressly 'forbidden' by law.¹⁰⁶

Considering how there are existing practices of data monetisation in terms of selling personal data to marketing agencies and advertisers by businesses¹⁰⁷ and how the Indian Government is considering to monetise public non-personal data,¹⁰⁸ it is evident that personal data monetisation is a lawful purpose because it is not prohibited expressly by any law in India.

2. There are several incentives for businesses in India to implement this personal data monetisation model. First, in the status quo, there is a limited free flow of personal data, because consumers may choose not to share their own personal data if they are able to access the same service using the personal data of family members or friends. For example, to open a four-screen Netflix account, the personal data of only one person is required, allowing the other three members to retain their personal data and still access the service. Further, consumers may share only limited attributes of their personal data such as e-mail addresses or phone numbers due to increased possibility and exposure to marketing and promotional communications.¹⁰⁹

Now, however, the personal data monetisation model allows an increased free flow of personal data. This is because it gives an incentive to consumers to share and sell their personal data to earn money. For example, in the case of a four-screen Netflix account, all four individuals will be incentivised to share their personal data with Netflix to earn money. This increased free flow benefits businesses because when personal data is sold to them, it increases their ability to utilise such data for various purposes, such as analysing the tastes and preferences of consumers, predicting consumer trends and patterns and earning more revenue

by selling on this personal data to third parties such as advertisers and marketing agencies.¹¹⁰

Secondly, in the status quo, the possibility of data privacy and security litigations and disputes becomes high because: (a) consumers are becoming increasingly aware of their rights in their personal data¹¹¹ owing to increased data protection under newly introduced domestic laws such as the DPDP Act; and (b) consumers are becoming increasingly concerned about multiple mass data breaches and misuse of personal data.¹¹² The aforementioned two reasons, ie increased consumer awareness and concern about personal data, are due to the fact that consumers retain ownership and interest in their personal data. Further, in the worst-case scenario, if such personal data is leaked or misused by businesses, consumers are less likely to ignore this, as there are no 'ignoring incentives' to do so. Under the personal data monetisation model, however, in the event of a data breach or misuse of personal data by businesses, consumers are more likely to ignore it. Consequently, the possibility of data privacy and security litigations and disputes is reduced for the businesses because: (a) consumers would be licensing their 'interest' in their personal data to the businesses, which would grant the businesses increased autonomy and flexibility regarding the utilisation of data for increasingly different purposes; and (b) consumers would be getting paid to sell their personal data to businesses, resulting in them comparing their priority of earning money with data breach/misuse and typically choosing the former over the latter. This inherent self-centric nature of consumers is likely to better benefit the businesses.¹¹³

Thirdly, tax benefits exist for the Government to implement the personal data monetisation model because the authors' personal data monetisation model

leads to an increase in the disposable income of individuals, which, in turn, leads to an increased capability for the central and state governments to collect more taxes on the increased expenditure of individuals. These tax benefits may be in the form of subsidies and exemptions for businesses to process and utilise more personal data. In other words, businesses and the Government would profit from such tax benefits.

3. Incentives for consumers to agree to the implementation of the personal data monetisation model are twofold. First, it increases the ability of consumers to earn more money. Considering how the unemployment rates have been rising over the past few years in India,¹¹⁴ consumers are likely to agree to the personal data monetisation model to meet their needs and earn a sustained living. Secondly, it increases the disposable income of consumers, who are likely to spend this income on luxury goods and services as well as necessities.¹¹⁵ This is likely to increase overall consumer satisfaction and happiness levels due to their ability to meet their needs, earn a sustained living and obtain luxury.
4. Apart from the stakeholder-specific incentives, there exist several general incentives that are likely to fuel acceptance from the Indian Government, businesses and consumers. These are primarily in terms of obligations imposed on the data principals under Section 15 of the DPDP Act as follows.¹¹⁶ First, the data principal is not allowed to impersonate any other person while giving personal data for a specific purpose.¹¹⁷ In the context of personal data monetisation, it ensures there is no identity theft of consumers and prevents a data principal from unlawfully gaining from the personal data of another data principal.

Secondly, the data principal is required to furnish their authentic information

to the data fiduciary, which is capable of being verified by the data fiduciary.¹¹⁸ In the context of personal data monetisation, it ensures that there is no falsified or inauthentic personal data given by the data principal to the data fiduciary. It also reduces the possibility of a data principal giving different personal data for the same attributes. Suppose a data principal is required to provide their e-mail address (personal data) for the purpose of collecting e-mails (attributes) from the data fiduciary. Now, the data principal will be able to give only those e-mail addresses that are registered with an authentic service like Gmail or Yahoo and earn money for only such authentic, verifiable personal data (e-mail address) given for the same attribute (e-mails). This prevents the data principal from unfairly and unlawfully earning a disproportionate amount of money by giving false and unverifiable personal data (unregistered and fake e-mail addresses) for the same attribute (e-mails).

Thirdly, it ensures that the data principal does not suppress any material information regarding any of their personal data, unique identifier, proof of identity or proof of address.¹¹⁹ In the context of personal data monetisation, it ensures that the data principal does not unlawfully earn money and profit by suppressing material information regarding their personal data. For example, if the data principal furnishes a residential address and uses a passport as proof of address, they may also be required to disclose material information regarding the passport, such as its validity. This is done to ensure that the data principal does not unlawfully earn money by selling personal data based on an invalid document such as an expired passport.

Thus, these obligations of the data principal under the DPDP Act assist the stakeholders in the context of personal data monetisation in a threefold manner:

(a) from the businesses' perspective, it reduces the possibility of litigation and disputes for businesses as they would be transacting with consumers based on lawful personal data given in a lawful manner; (b) from the consumers' perspective, it ensures that consumers' personal data and their ability to earn money from its sale is secured as it reduces the possibility of unlawful gains and data theft; and (c) from the Government's perspective, it increases the sanctity of the personal data monetisation model as it helps to maintain law and order by reducing the possibility of identity thefts, unlawful gains and widening of the income disparity.

All the incentives mentioned above are bolstered by the fact that in the event of any conflict in the personal data monetisation model, this can be resolved in accordance with the dispute resolution procedure given in the DPDP Act. For example, if there is a conflict between the data principal and the data fiduciary, either can approach the Data Protection Board of India (DPBI) under Section 27 of the DPDP Act¹²⁰ and follow the procedure laid out in Section 28 of the Act to resolve the conflict.¹²¹ In the event that either party is unsatisfied with the decision of the DPBI, they can appeal to the Appellate Tribunal under Section 29 of the Act,¹²² which will be the 'Telecom Disputes Settlement and Appellate Tribunal'.¹²³ Both the DPBI and the Appellate Tribunal are vested with the powers of a civil court under Sections 28(7)¹²⁴ and 30(1) of the Act,¹²⁵ respectively. Additionally, both the DPBI and the Appellate Tribunal are empowered to impose monetary penalties under Section 33(1) of the Act, ranging from INR 50 crores to INR 250 crores.¹²⁶ Thus, the incentives mentioned above help create a conducive environment for the smooth implementation of the personal data monetisation model in India.

OPERATIONALISING THE PERSONAL DATA MONETISATION MODEL IN INDIA: PROCESS AND DIGITAL INFRASTRUCTURE

On a brief perusal of the DPDP Act it is evident that it is substantive in nature because it primarily recognises the substantive elements as follows: (a) the need for express consent;¹²⁷ (b) the legitimate uses for which personal data may be processed;¹²⁸ (c) the general obligations of data fiduciaries;¹²⁹ and (d) the general rights and duties of data principals.¹³⁰ The Act, however, fails to discuss the operative elements as follows: (a) how such consent would be materialised in practice and be managed by consent managers; (b) how the materialisation of various legitimate uses for processing personal data will be regulated to maintain compliance with the DPDP Act; (c) how small to large-scale businesses would be expected to comply with their general obligations of data fiduciaries; and (d) how the rights and duties of data principals would be enforced in day-to-day business and consumer activities. Such operative elements need to be dealt with by the Indian Government because: (a) these operative elements are derived from the substantive elements in the DPDP Act; and (b) the operative elements would affect the day-to-day business and consumer activities. Thus, operative elements may be introduced under the DPDP Act in the form of separate Draft DPDP Rules,¹³¹ provided such operative elements are derived from the substantive elements of the DPDP Act.

The recently released Draft DPDP Rules provide a framework to address these operational gaps by detailing the processes necessary to implement the DPDP Act effectively. For instance, Rule 5 specifies the standards for obtaining express and informed consent.¹³² This rule mandates that data fiduciaries provide granular consent options, ensuring transparency in how personal data is used.¹³³ Additionally, Rule 6 outlines the mechanisms for withdrawal of consent,

requiring that such withdrawal be as simple as the process of giving consent.¹³⁴ These provisions are critical for operationalising the personal data monetisation model as they ensure that data principals retain control over their data throughout the transaction life cycle.

Similarly, in the context of personal data monetisation, a personal data monetisation model can be introduced under the DPDP Act because: (a) the model is an operative element since it will operationalise how personal data will be bought and sold; and (b) the model, as an operative element, is derived from the substantive elements of the DPDP Act, as this Act substantively discusses three key substantive elements which would be present as key operational players in the model. First, it is the 'data principal', defined as any individual to whom the personal data relates.¹³⁵ Secondly, it is the 'consent manager', defined as a single point of contact enabling the data principal to give, manage, review and withdraw consent.¹³⁶ The consent manager allows the same through an interoperable and accessible platform that is transparent in its working and nature.¹³⁷ This ensures that the data principal has the ability to freely withdraw consent in compliance with Section 6(4) of the DPDP Act.¹³⁸ Thirdly, it is the 'data fiduciary', defined as a person responsible for processing personal data and deciding its purpose and means of processing.¹³⁹ Thus, these three key substantive elements of the DPDP Act become key operational players in the personal data monetisation model, whereby these operative elements may be introduced in the form of separate operationalisation Draft DPDP Rules framed under the DPDP Act.

After having dealt with how the personal data monetisation model can be introduced in India, it is imperative to discuss how this model will function. It will be operational in the form of a 'data trust'. The data principal will be the 'beneficiary' in the data trust because it is their personal data that is being managed and handled for their benefit by

the consent manager. The data principal will also be the 'settlor' as they transfer their assets (personal data) into the trust (data trust) for the purpose of selling it (personal data monetisation). The consent manager will be the 'trustee' in the data trust because they will be responsible for managing the consent preferences and authorisations of the data principal by giving, managing, reviewing and withdrawing consent as may be permitted by the data principal. The data fiduciary will be the 'beneficiary' in the data trust because the data fiduciary benefits from the personal data being sold to them by the data principal. These benefits include the usage of personal data for marketing, internal research and sharing with third parties by the data fiduciary to earn more revenue.

This means that:

- a. The data principal is likely to be individuals such as consumers and not companies, limited liability partnerships (LLPs), Hindu Undivided Families, body corporations, Association of Persons, Body of Individuals and any other similar

organisation as may be defined as a 'person' under Indian law.

- b. The data fiduciary is likely to be any 'person' (and not individuals such as consumers) such as companies, LLPs, Hindu Undivided Families, body corporations, Association of Persons, Body of Individuals and any other similar organisation as may be defined as a 'person' under Indian law such as the Indian Companies Act, 2013, Indian Income Tax, 1961, Indian Foreign Exchange Management Act, 1999 or any other relevant Indian law as may be applicable, whereby the organisation intends to provide services to the data principals in exchange for personal data and is not expressly prohibited by law to do the same.
- c. The consent manager will be a governmental body of persons comprising hired and skilled personnel.

Thus, this leads to the creation of three integral elements in a trust, which, in turn, leads to the creation of a data trust (see Figure 1).

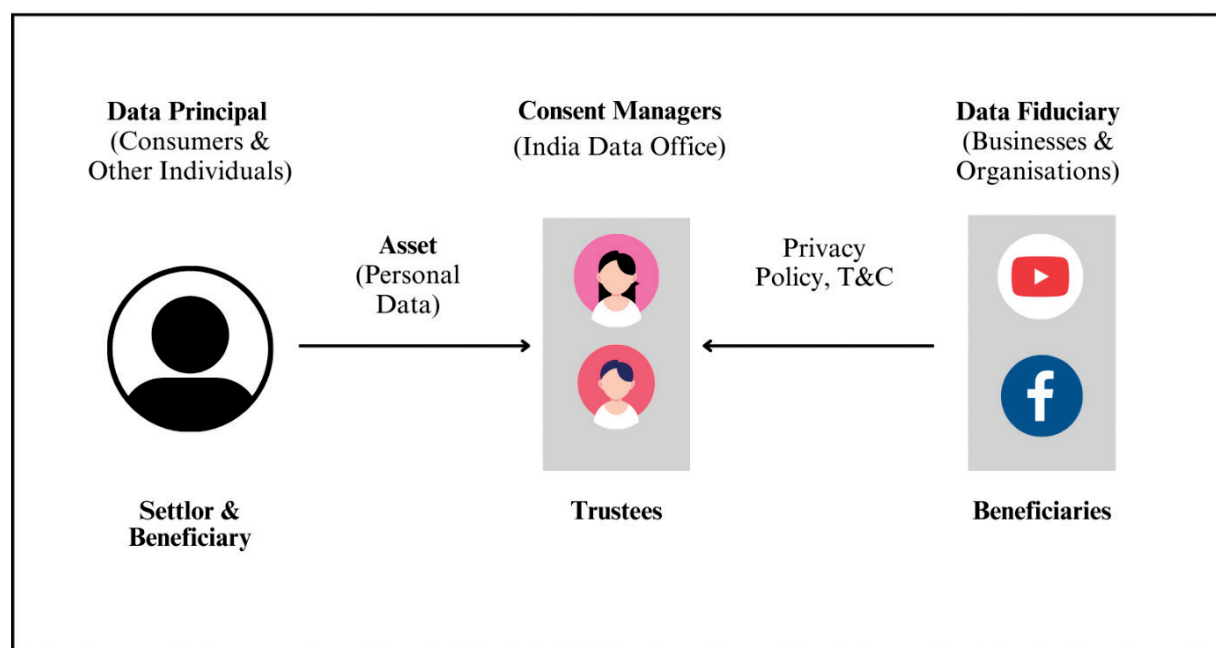


Figure 1: The data trust for the personal data monetisation model in India

The data trust will operate on a similar concept of licensing, as mentioned earlier for the Indian copyrights framework, for the purpose of monetisation and selling of personal data as an economic commodity or asset (see Figure 2). This licensing of personal data will work in the following manner:

1. First, the data principal will grant 'interest' in their personal data to the data fiduciary through a consent manager. This means that the data principal will provide their personal data and consent preferences and authorisations to process their personal data to the consent manager, who acts as an intermediary. This will later be transferred to and shared with the data fiduciary. This leads to the data principal being in control and ownership of the volume and nature of their personal data that can be processed by the data fiduciary. This control and ownership exists because of the assistance and facilitation provided by the consent manager. Thus, the data principals sell their personal data by licensing
- it to the data fiduciary through the consent manager in return for monetary compensation.
2. Secondly, the consent manager will transfer the data principal's personal data to the data fiduciary in accordance with the consent preferences and authorisations of the data principal. The data fiduciary can then utilise the data for permitted purposes under the DPDP Act.
3. Thirdly, after the data fiduciary receives the personal data, the data fiduciary compensates the data principal directly for their personal data, either through a lump sum payment or regular royalties. The terms of payment are governed by a separate agreement between the data principal and the data fiduciary, which may include provisions for revenue sharing or fixed compensation.

The Draft DPDP Rules further facilitate this process by providing clarity on the obligations of data fiduciaries. Rule 9 mandates that significant data fiduciaries conduct periodic audits and

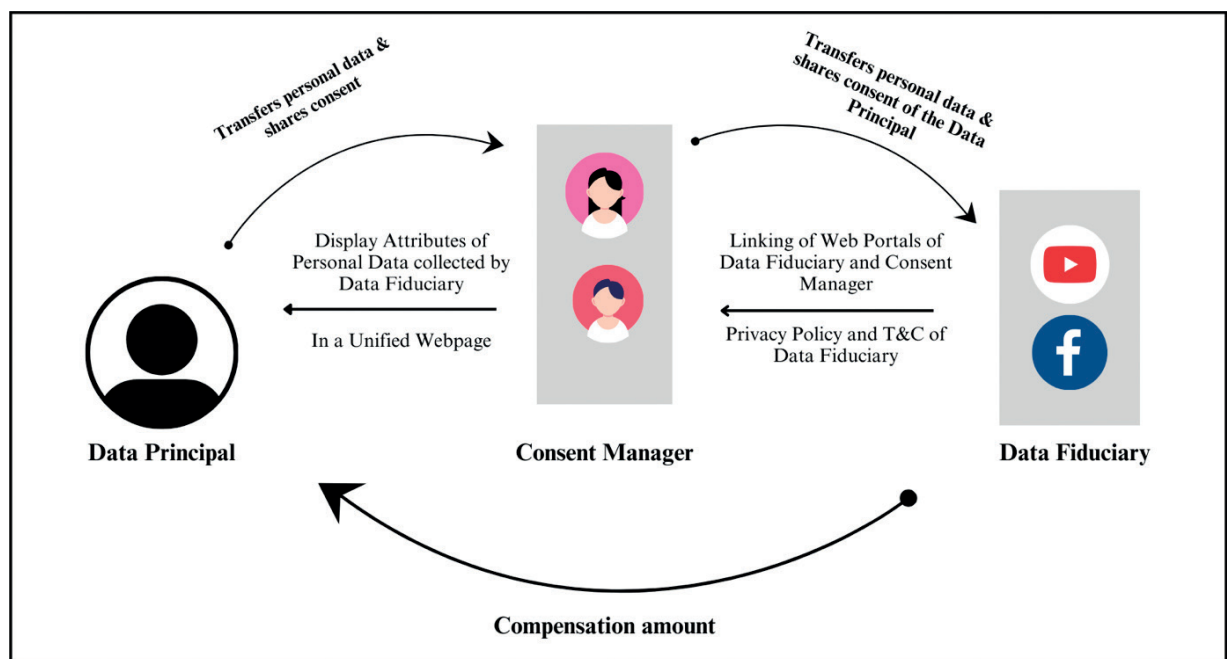


Figure 2: The operationalisation of the personal data monetisation model in India

submit compliance reports, ensuring that monetisation activities adhere to the principles of transparency and accountability.¹⁴⁰ Additionally, Rule 11 introduces the concept of grievance redressal mechanisms, which can address disputes arising from data monetisation agreements.¹⁴¹ These provisions ensure that the operational framework for personal data monetisation is robust and capable of safeguarding the rights of data principals.

From the mechanism mentioned above, it is evident that a unique digital infrastructure is required on which the personal data monetisation model can operate. To resolve this conundrum, the personal data monetisation model can utilise the digital infrastructure provided by India Stack. In simple terms, India Stack is a set of application programming interfaces (APIs)¹⁴² — rules and protocols that allow different software applications to communicate with each other. These APIs act as an intermediary by allowing one software application to use the functions and data of another. This unique functionality given by India Stack creates a unique, interoperable digital infrastructure for consumers, governments, businesses and start-ups to seamlessly share data with each other. It operates as follows with the help of its three layers:

1. *Identity layer*: In this layer, India Stack primarily utilises the Aadhaar system, India's national identity programme.¹⁴³ This layer allows users to perform the following functions on a cross-platform basis (between two or more different platforms): remotely authenticate various attributes (eg name, age, mobile number, e-mail address), exchange digitally signed and universally accepted copies of records (eg driving licence, insurance policies, education diplomas) and sign documents by utilising a government-backed electronic signature service.¹⁴⁴
 - *Illustration*: A consumer wants to open a new bank account. Instead of physically visiting the bank's nearest branch with various identification documents, the consumer may use their Aadhaar number. In this case, the bank can use the Aadhaar API to verify the consumer's identity online quickly and securely. This electronic verification through Aadhaar simplifies the process, making it quicker, effective, efficient and more convenient for both the consumer and the bank.
2. *Payments layer*: In this layer, India Stack utilises the unified payments interface (UPI) developed by the National Payments Corporation of India (NPCI).¹⁴⁵ This layer allows consumers and businesses to pay for goods and services through UPI,¹⁴⁶ a payments markup language that runs on a 'central switch' operated by NPCI. A single central NPCI server is connected with licensed banks in India,¹⁴⁷ which sends payment messages back and forth between the banks.¹⁴⁸ Thus, the UPI system is three-tiered. The first tier is the digital infrastructure provided by NPCI in the form of a single server.¹⁴⁹ The second tier is the licensed banks.¹⁵⁰ The third tier is the 'fintech layer', through which third-party payment applications such as Google Pay, PayTM and PhonePe access the other two tiers for the purpose of payments.¹⁵¹ The reason why UPI has gained extreme popularity in India can be attributed to the fact that it is interoperable (as UPI has an API), and transactions can be done through UPI with the help of third-party payment applications without any additional costs.¹⁵²
 - *Illustration*: A consumer wants to pay for their groceries at a local store. Instead of using cash, they can now open a UPI-enabled third payment app on their smartphone, enter the merchant's UPI ID, and transfer the amount directly from their bank account to the merchant's account through the UPI-enabled app. The

consumer can do the same at a different local store (having a different bank) as UPI is interoperable, ie the consumer can use the same UPI-enabled app to make payments to different merchants having different banks. The transactions are quick and secure and do not require physical cash or a card.

3. *Data layer*: In this layer, India Stack works on the Data Empowerment and Protection Architecture (DEPA), where documents are digitally stored, accessed and exchanged by utilising secure data exchange platforms such as DigiLocker, where personal data and documents are securely exchanged and verified.¹⁵³ Further, this layer primarily operates with the help of regulated entities known as consent managers, who facilitate the sharing and consuming of data between the consumer and third parties, as per the consent preferences and authorisation given by the consumer.¹⁵⁴

- *Illustration*: A consumer is applying for a loan. The lending institution requires the consumer's income and address proof. Instead of submitting physical documents, the consumer can utilise the services of DigiLocker, whereby the consumer gives consent through the DigiLocker API (consent manager) to the lender to access the consumer's securely stored online digital documents, such as income tax return or utility bill. This is an effective and efficient process that reduces the risk of document loss and forgery.

In the context of the personal data monetisation model, the India Stack digital infrastructure can be deployed as follows:

1. First, by utilising the identity layer, the data principal can verify their personal data for various attributes such as name, age, e-mail address and phone number and give consent by electronically signing documents of data fiduciaries,

such as the privacy policy and terms and conditions that may exist in the form of clickwrap agreements. This verification can be done with the help of the data principal's Aadhaar number, where the verification may be performed by the consent manager at their web portal, which will be linked with the Aadhaar API. The consent manager's web portal will be connected to the data fiduciary's web portal to allow the latter to detect different data principals who have given consent to the data fiduciary to process different types of personal data.

The data will be processed by the data fiduciary and will be visible in the form of a simplified and unified table on the consent manager's web portal (see Figure 3). This table will be created automatically by interlinking between the consent manager's web portal and the data fiduciary's web portal. It will be created based on the terms and conditions, cookies policy, privacy policy, end-user licence agreement (if any) and any other relevant policy of the data fiduciary. This will allow the data principal to make an informed decision and give better-informed consent selectively to various data fiduciaries for the purpose of personal data monetisation. It is noteworthy to mention that the relationship between the data principal and the data fiduciary for selling personal data by way of licensing and receiving monetary compensation is likely to be governed by the various platform policies of the data fiduciary and any other agreement as may be expressly entered into between them.

The reason why this unified consent mechanism is different from the existing mechanism in the status quo is that under the DPDP Act, the data principals would be required to give consent individually, independently and separately to different data fiduciaries. Now, however, in the new unified consent mechanism

mentioned above under the personal data monetisation model, the data principals can give and manage their consent from a unified web portal for all data fiduciaries.

For example, in the existing mechanism under the DPDP Act, the consumer would be required to give separate consent to Spotify, Apple Music and SoundCloud on their individual platforms/web portals/apps to access their services. Now, however, with the new unified mechanism under the personal data monetisation model, the consumer would be able to give and manage consent for the same services on the unified web portal of the consent manager. This web portal will display a simplified and unified table that will show different types of personal data being collected by Spotify, Apple Music and SoundCloud in a comparative manner. The table will be based on the terms and conditions, cookies policy, privacy policy, end user licence agreement (if any) and any other relevant policy that will allow the consumer to make better-informed decisions and give better-informed consent to any of the services (Spotify, Apple Music, SoundCloud) to process their personal data for monetisation.

This table ensures that the data principal is notified as to what types of personal data are collected by the data fiduciary and for what purposes it will be processed by being in compliance with Section 5(1) of the DPDP Act.¹⁵⁵ Further, the table ensures that consent is given by the data principal in a 'free, specific, informed, unconditional and unambiguous with a clear affirmative action' manner, agreeing to the processing of their personal data for the specified purpose by the data fiduciary.¹⁵⁶ This ensures compliance with Section 6(1) of the DPDP Act, which mandates the aforementioned.¹⁵⁷ It is imperative to note that consumers may have an option of giving only certain types of personal

data to a data fiduciary on the web portal of the consent manager; however, this option will be completely contingent on the policies of the data fiduciary concerned.

Additionally, as this table is essentially a request for obtaining express consent from the data principal, it should be displayed to the data principal in clear and plain language. The data principal will be allowed to give consent in English, and the table must give the details of the data fiduciary's DPO or any other authorised personnel of the data fiduciary to allow the data principal to exercise their rights under the DPDP Act.¹⁵⁸ This must be done to ensure compliance with Section 6(3) of the DPDP Act which mandates the aforementioned.¹⁵⁹

At this point in time, a query arises: what will be the basis or metric on which different personal data attributes from different platforms will be grouped together in the form of a simplified and unified table and displayed to the data principals? The answer is that different personal data attributes from similar platforms will be grouped together and displayed in different categories based on services such as music, cartoon, anime, educational, etc. This unification in a comparative manner would ensure better-informed consent by the data principal.

2. Secondly, by utilising the payments layer, the data principal and the data fiduciary can enter into a separate agreement for payment purposes under the personal data monetisation model. This means that when the data principal has sold their personal data to the data fiduciary by way of licensing, the data principal is bound to receive monetary compensation from the data fiduciary. This may be in the form of a clickwrap agreement that can be displayed in a simplified manner on the consent manager's web portal at the time when the data principal is giving and managing their consent. The reason

why the consent manager's web portal would be able to display this clickwrap monetisation agreement is that it will be privately placed on the data fiduciary's web portal and interlinked with the consent manager's web portal. It would be capable of being viewed and accessed by the data principal, consent manager and data fiduciary only. Further, the data principal would directly receive monetary compensation from the data fiduciary through any of the UPI-enabled third-party payment apps. The consent manager will not be involved in the disbursement of this monetary compensation. The limited role that the consent manager will play in the payments layer is that of displaying the payment clickwrap agreement in a simplified manner.

This monetary compensation may be either (a) a one-time lump sum that the data fiduciary may pay out to the data principal based on the percentage of the revenue earned from using the data principal's personal data for a certain period, or a pre-agreed, fixed lump sum; or (b) a regular payment mechanism (weekly, monthly, bi-monthly, quarterly, semi-annually, annually or any other frequency as may be agreed) by way of royalties based on the percentage of the revenue earned from using the data principal's personal data for a certain period, or a pre-agreed, fixed amount paid at regular intervals.

The reason why the new payment mechanism is different from the existing payment mechanism under the DPDP Act is that, in the existing payment mechanism, the data principals do not sell their personal data to the data fiduciary, they merely give their consent to process their personal data to the data fiduciary to purchase goods or access services. Now, however, under the personal data monetisation model, the data principals are going a step beyond merely giving consent, they are selling their personal

data to the data fiduciary in return for monetary compensation. For example, in the existing payment mechanism in the status quo under the DPDP Act, the consumer would give their consent to Spotify to collect and process their personal data to allow them to listen to music. Now, however, with the new payment mechanism under the personal data monetisation model, the consumer would be able to (a) give their consent to Spotify to collect and process their personal data; and (b) receive monetary payment (lump sum or royalties as may be agreed) for selling their personal data to Spotify by consenting to its collection and processing.

At this point, a query might arise whether a data fiduciary can include an exclusivity clause in the clickwrap monetisation agreement or any of its other policies between the data fiduciary and the data principal for various purposes as follows: (a) to reduce the possibility of the data principal selling their personal data to more than one data fiduciary, ie reduction of competition for the data fiduciary; and (b) to ensure that the data principal's unique personal data is exclusively utilised by the data fiduciary. This can have detrimental effects on the data principal because it restricts the free flow of personal data in the digital sphere and the ability of the data principal to earn more money.

Although the possibility of data fiduciaries imposing exclusivity clauses exists, it is imperative to note that such clauses will fall under the ambit of Section 27 of the Indian Contracts Act, 1872 (1872 Act).¹⁶⁰ Section 27 essentially states that if any agreement restrains anyone from carrying out a lawful business, then such agreement is void.¹⁶¹ Further, in *Niranjan Shankar Golikari v The Century Spinning and Manufacturing Company Limited*,¹⁶² the Supreme Court held that if an agreement is unreasonable

or one-sided, it is in restraint of trade or business and void under Section 27. Similarly, in the context of personal data monetisation, if a data fiduciary imposes such an exclusivity clause, it will be unreasonable and one-sided because it restricts the free flow of personal data in the digital sphere and limits the earning ability of the data principal. Thus, data fiduciaries are less likely to impose such an exclusivity clause under the personal data monetisation model, as it is likely to be rendered void by the Indian courts.

3. Thirdly, by utilising the data layer, platforms such as DigiLocker, governed by the DEPA, can enable the secure storage and exchange of personal data, managed by regulated consent managers. The consent managers will be the body of persons responsible for giving, managing, reviewing and withdrawing consent from the data principal to the data fiduciary. In other words, the consent manager will act as an intermediary between the data principal and the data fiduciary for the following purposes: (a) to allow the data principal to purchase goods or access services of the data fiduciary by managing the data principal's consent and sharing it with the data fiduciary; and (b) to allow the data fiduciary to earn revenue from the data principal's personal data and pay monetary compensation to the data principal for personal data monetisation.

Further, the consent manager's web portal will be a secured portal where various documents would be securely stored and exchanged, such as electronic signatures of the data principal; terms and conditions, cookies policy, privacy policy and any other policy of the data fiduciary; and any other documents as may be exchanged between the data principal and the data fiduciary for the twin purposes mentioned above. The operability of the consent manager will be the same as previously described:

- a. The data principal will transfer their personal data and share their consent preferences and authorisations to the consent manager based on the simplified and unified table showing the types of personal data being collected by the data fiduciary.
- b. The consent manager will transfer the data principal's personal data and share their consent preferences and authorisations with the data fiduciary by virtue of the fact that the web portals of the consent manager and the data fiduciary are interlinked.
- c. The data fiduciary will provide services to the data principal as per their consent preferences and authorisations.
- d. The data fiduciary will monetise the data principal's personal data by utilising it for various purposes to earn revenue.
- e. The data fiduciary will pay monetary compensation to the data principal for selling personal data by way of licensing.

Additionally, it is noteworthy to mention that consent managers are an appropriate choice to manage the consent preferences and choices for the data principals' personal data because individuals' ability to give and manage free and informed consent is obstructed by their behaviour patterns, as explained earlier, in terms of bounded rationality, heuristic familiarity tendency and default inertia.¹⁶³ Due to this, consent must be managed and facilitated by neutral and independent bodies such as consent managers.

Further, it is imperative to discuss the situation when the data principal decides to withdraw consent. In such a scenario, the following is likely to happen: (a) the amount accrued until the date and time of consent withdrawal and the amount already due and outstanding will be paid to the data principal as per the clickwrap monetisation

Consent Manager
District of Kollam

Attributes of Personal Data Collected					
Name	✓	✓	✓	✓	✓
Phone Number	✓	✓	✓		
Email ID	✓	✓	✓		
Address	✓	✓			

Figure 3: The simplified and unified personal data table visible on the consent manager's web portal

agreement signed between the data principal and the data fiduciary; and (b) the data fiduciary will delete the personal data of the data principal in compliance with Section 8(7) of the DPDP Act, which states that the data fiduciary must erase the personal data as and when the data principal withdraws their consent.¹⁶⁴ As the data fiduciary is likely to enter into data processing agreements (DPAs) with third parties such as advertisers and marketing agencies,¹⁶⁵ the third parties would also be bound to comply with the requirements of Section 8(7) of the DPDP Act, as similar or same obligations are likely to be incorporated in the DPAs.

The aforementioned discussion is primarily contingent on the assumption that the data principal is not deceased. Due to this, the question arises whether the data principal's personal data will still be monetised in the event of their death. The answer is that as long as the data principal did not expressly agree to withdraw consent regarding processing their personal data after their death, the personal data will continue to be processed and monetised. The amount realised by the deceased data principal can be

given to authorised individuals (eg assigns, legal heirs or authorised representatives) as may be agreed between the data principal and the data fiduciary before the data principal's death.

The proposed operational framework addresses several key concerns that may be raised by critics of the personal data monetisation model:

- To mitigate the risks of exploitation in the context of poverty and low privacy awareness, stringent safeguards will be implemented to ensure that consent is truly informed and voluntary. Neutral consent managers, as independent entities, will be appointed and will play a critical role in addressing issues of bounded rationality and default inertia among data principals.
- The model will include mechanisms to evaluate the fair value of data, considering its context-dependent and evolving nature. A transparent pricing framework, overseen by regulatory authorities, will be adopted to standardise compensation and prevent exploitation.

- Data principals will be allowed to retain control over their data even after monetisation. This will be facilitated via the Draft DPDP Rules that explicitly specify that licensing agreements do not waive the data principal's fundamental right to privacy or their ability to revoke consent. Additionally, Section 8(7) of the DPDP Act will be strictly enforced to ensure that data fiduciaries delete personal data upon consent withdrawal.
- The potential for data misuse by downstream entities will be curtailed through strict compliance requirements and accountability mechanisms for data fiduciaries and third-party processors. This will be facilitated via DPAs that will include clauses mandating compliance with the DPDP Act and the operationalisation Draft DPDP Rules.

INSTITUTIONAL FRAMEWORK FOR PERSONAL DATA MONETISATION IN INDIA

After having discussed the need for personal data monetisation and how it can be implemented through a data trust by utilising the digital infrastructure of India Stack, it is imperative to consider how the institutional framework of the data trust will be established and operationalised. This can happen through the India Data Office (IDO) under the 2022 Policy introduced by the Ministry of Electronics and Information Technology (MeitY) (coupled with the DPDP Act).¹⁶⁶ Before discussing the composition of the IDO in the context of personal data monetisation, it is essential to review the 2022 Policy. The 2022 Policy seeks to replace the National Data Sharing and Accessibility Policy (NDSAP) and the Open Government Data Platform.¹⁶⁷ The NDSAP is a policy implemented in 2012 in India that aims to allow the sharing of government-generated and government-owned non-sensitive data for the purpose of national planning and development.¹⁶⁸ The

OGDP was established under the NDSAP and is essentially a one-stop platform to make government-owned shareable data available to the public in an easily accessible format.¹⁶⁹ Now, however, the NDSAP and OGD might be replaced by the 2022 Policy, which aims to share non-personal data 'created/generated/collected/archived by the Government of India directly or through authorized agencies by various Ministries/Departments/Organizations/Agencies and autonomous bodies'.¹⁷⁰

In the context of personal data monetisation in India, the 2022 Policy can be utilised (coupled with the DPDP Act) because the 2022 Policy is still in the draft stage. This means that it can be amended to suit the personal data monetisation model's needs and requirements accordingly. One of the main amendments would be to expand the model's scope beyond governmental authorities and non-personal data by making the 2022 Policy applicable to the following personal data: (a) personal data that is created and generated by any 'person' (such as consumers and other individuals and not companies or organisations); and (b) personal data that is collected, processed, stored or archived by private companies, businesses and organisations. This amendment would ensure that the commercial value of the personal data that is created and generated by consumers directly accrues to them and not from the governmental authorities to private organisations. Further, the 2022 Policy must be amended to make it applicable to the personal data that is collected, processed, stored or archived by the central and state governments; however, it may continue to apply to non-personal data as MeitY deems fit and appropriate.

The 2022 Policy discusses the establishment of an IDO, which is tasked with the responsibility of implementing the 2022 Policy at various levels and to 'streamline and consolidate data access and sharing of non-personal data repositories'.¹⁷¹ This aforementioned objective and function

of IDO can be expanded to personal data to suit the needs and requirements of the institutional framework for personal data monetisation in India. In other words, the Indian Government has already proposed a national institution (IDO) for streamlining and consolidating personal data which can be utilised to institutionalise and operationalise the data trust and the personal data monetisation model by enabling data processing in a structured manner, as outlined below. Under the ambit of IDO, consent managers (an element of the data trust) can be established in the following manner:

1. The district-level consent manager will handle the operational aspects of data processing, ie collecting, recording, sharing and storing. For instance, when a data principal consents to share or monetise their data with a data fiduciary, the district-level consent manager processes this transaction and ensures compliance with the data principal's terms. In other words, if the data principal has expressly given their consent to process their personal data and sell it to certain data fiduciaries, only these data fiduciaries will be able to access the personal data of the data principal as shared and transferred through the district-level consent manager. The district-level consent manager's web portal will be accessible by the consent managers at all three levels (national, state and district). Primarily, however, it would be utilised by the district-level consent manager as it is directly and actively involved in the activity of processing the data principals' personal data. This web portal of the district-level consent manager will be connected to the data fiduciary's web portal. Due to this, the attributes for which personal data is being collected by the data fiduciary will be displayed to the data principals in the form of a simplified and unified table

on the district-level consent manager's web portal. The consent preferences and choices and personal data of the data principals are stored and maintained on the servers of the district-level consent manager to facilitate sharing and transfer to the data fiduciary, until the data principal withdraws their consent to disallow the district-level consent manager from doing the same. This means that the data principal can withdraw consent from two elements: (a) from the data fiduciary in accordance with Section 8(7) of the DPDP Act, which states that the data fiduciary will not retain the personal data and erase it when the data principal expressly withdraws consent;¹⁷² and (b) from the consent manager.

The servers of the district-level consent manager will be connected to the state-level and national-level consent managers, and the personal data collected in these servers will be stored in a single, unified national platform such as the OGD. This means that the storing of personal data will be done by the district-level consent manager in platform to which the state-level and national-level consent managers will have access as and when allowed under Section 7(b)¹⁷³ and any other relevant provisions of the DPDP Act. There will be one district-level consent manager in each district, and each district-level consent manager will have a physical office.

It is imperative to note that the consent preferences and choices and personal data of data principals will be stored and maintained in the data servers of only one district-level consent manager based in the district with the highest population. This is because districts with high population levels require and possess relatively higher security and technology infrastructure as they become industrialised, urbanised and modernised relatively faster than less populated districts. For example, in

Telangana state, Hyderabad is the most developed district in terms of security and technology infrastructure as it has the highest population.¹⁷⁴ The same is true for other districts as well, such as Bengaluru (Silicon Valley of India) in Karnataka state.¹⁷⁵ Thus, storage and maintenance of consent preferences and choices and personal data of data principals in the district with the highest population ensures multiple technological and security benefits to the data principals, including efficient data processing, sharing and storage, faster data collection and less possibility of data leakage.

2. The state-level consent manager will be primarily responsible for checking the compliance of all the district-level consent managers with the local and the respective state's data privacy, security and protection laws. The state-level consent manager will do this by accessing the single, unified OGDG-like platform on which the personal data has been stored by the district-level consent managers. In case of non-compliance, the state-level consent manager will be empowered to order the concerned district-level consent manager(s) to rectify the non-compliance. In case of continued non-compliance despite the rectification order, the state-level consent manager will be empowered to impose a reasonable penalty on the concerned district-level consent manager(s), which can have a long-term deterrent effect. The state-level consent manager will be established with a physical office in the district with the highest population. This means that there would be two consent managers in the district with the highest population, i.e. a district-level consent manager and a state-level consent manager.
3. The national-level consent manager will function as the apex body responsible for ensuring that all consent managers operate within the legal framework of the DPDP Act. They will primarily be

responsible for checking the compliance of the district-level consent managers with the national, state and local laws for data privacy, security and protection. This will be done by accessing the single, unified national OGDG-like platform. Further, the national-level consent manager will be responsible for maintaining this platform to ensure the deployment of state-of-the-art security protocols, data processing standards and technological infrastructure. This will ensure a streamlined flow of personal data across all levels while mitigating risks of unauthorised access or breaches. There will be a single national-level consent manager.

4. Consent manager roles at all levels will be staffed by personnel with expertise in data protection and privacy. Additionally, the institutional framework should include mechanisms for training and capacity building to address emerging privacy challenges, especially with advancements in AI and big data analytics. The composition of the personnel in the body of the consent manager at the national, state and district levels will be at the discretion of India's central and state governments.

To address the issue of pricing personal data, the IDO can establish a robust and transparent valuation mechanism that reflects the diverse and evolving nature of personal data. Unlike tangible goods, the value of personal data is highly context-dependent, varying according to factors such as type of data, its potential for monetisation and its intended use. For instance, a data principal's browsing history may hold significant value for targeted advertising but far less for medical research, whereas health records could command a higher premium in contexts involving clinical trials or pharmaceutical development. By adopting a contextual valuation approach, the IDO can ensure that data principals

are fairly compensated in line with the actual utility and market demand for their data. Moreover, the valuation mechanism must incorporate safeguards to prevent discriminatory practices, such as offering higher compensation for data from affluent individuals while undervaluing data from economically disadvantaged populations. This would align with the DPDP Act's principle of inclusivity and fairness.

Furthermore, the DPDP Act under Section 5(1) mandates that data fiduciaries issue detailed consent notices to inform data principals about how their data will be processed and monetised.¹⁷⁶ This requirement forms the backbone of the valuation mechanism, ensuring transparency and enabling data principals to make informed decisions about whether to participate in monetisation. By including specifics about how data will be utilised, the potential revenue it may generate and the terms of the licensing arrangement, consent notices become not just a tool for compliance but a means of empowerment for data principals.

While establishing fair pricing mechanisms and transparent consent notices is crucial, the institutional framework must also address the overarching risks of surveillance capitalism, which thrives on exploiting personal data to maximise profits. The proposed framework must impose strict limitations on the powers of consent managers and data fiduciaries to prevent such exploitation. Licensing agreements for personal data must be carefully structured to prohibit fiduciaries from processing or sharing data beyond the scope explicitly agreed upon by the data principal. For instance, if a fiduciary licenses a data principal's location data for urban planning, the same data must not be repurposed for targeted advertising or shared with third-party advertisers without separate, explicit consent.

Section 8(7) of the DPDP Act reinforces this principle by requiring fiduciaries to erase personal data upon withdrawal of consent.¹⁷⁷ This provision ensures that data

principals retain control over their data even after it has been monetised. Withdrawal of consent must be facilitated through consent managers in a manner that is seamless, easily accessible and does not impose undue burdens on the data principal. For example, a centralised consent management portal could enable data principals to monitor which fiduciaries have access to their data and revoke access in real time, thereby eliminating the risks of unauthorised data retention or misuse.

By intertwining a fair pricing mechanism with stringent consent and usage limitations, the framework effectively counters the systemic risks posed by surveillance capitalism. Not only does this protect the fundamental right to privacy, but it also ensures that data monetisation operates as a tool for economic empowerment rather than exploitation. Together, the valuation mechanism, informed consent notices and strict regulatory oversight form a cohesive structure that balances the economic potential of personal data with the ethical imperative to safeguard individual autonomy and dignity.

Thus, the aforementioned allows the institutionalisation of the data trust and personal data monetisation model.

CONCLUSION

Considering the preceding arguments, the establishment of property rights over personal data is not only desirable but also essential to empower individuals in exercising greater autonomy, control and the right to informational self-determination. The current asymmetry in bargaining power between businesses and individuals results in significant inequities, where the commercial value of personal data overwhelmingly accrues to businesses. By recognising property rights in personal data, the personal data monetisation model provides an actionable framework to mitigate this power disparity, ensuring that data principals are

fairly compensated for their contributions to the digital economy.

The implementation of the personal data monetisation model, grounded in the DPDP Act, offers a market-oriented approach to address these inequities. While the DPDP Act focuses on protecting personal data through consent and fiduciary obligations, the monetisation model extends this framework by enabling individuals to license their data as a valuable economic commodity. This model is supported by a robust institutional and digital infrastructure, including the India Stack's identity, payments and data layers, alongside the establishment of consent managers under the operational framework of the IDO. These consent managers, based at district, state and national levels, ensure the secure and transparent management of consent, thereby addressing concerns about data misuse and ensuring compliance with privacy regulations.

The potential risks associated with this model, particularly the dangers of surveillance capitalism and privacy exploitation, necessitate robust safeguards. The institutional framework must prioritise informed consent, transparent data valuation mechanisms and stringent limitations on the processing and sharing of personal data. For instance, licensing agreements should explicitly restrict data fiduciaries from further sharing or repurposing personal data without renewed consent. The DPDP Act's provisions, such as Section 5(1) mandating detailed consent notices and Section 8(7) requiring data erasure upon consent withdrawal, provide a legal foundation to enforce these safeguards effectively.

The issue of pricing personal data remains critical, and the IDO's role in establishing valuation mechanisms is indispensable. By considering contextual factors such as data type, its monetisation potential and its intended use, the valuation framework can ensure fair compensation while preventing discriminatory practices. Furthermore, provisions must account for inferred data,

ensuring that individuals benefit not only from the raw data shared but also from its derivative uses in AI and big data analytics.

Finally, while data monetisation has the potential to empower individuals, it is imperative to recognise and address the digital divide in India. The model must remain voluntary, allowing individuals to opt out and continue accessing services without mandatory data-sharing requirements. Moreover, the risks of creating centralised repositories like India Stack must be mitigated through decentralised security measures to prevent the creation of vulnerable honeypots.

By addressing these concerns and leveraging the DPDP Act's regulatory framework, the personal data monetisation model can transform the data economy in India. It complements the DPDP Act's objectives by fostering a balance between economic empowerment and privacy protection, ensuring that individuals not only reclaim control over their personal data but also participate equitably in the economic value it generates. In an age where privacy is often viewed as a myth, this model reimagines it as a tangible, enforceable and economically significant right.

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